STA - 102

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Topic : Probability , Expectation of sum and product and variance of random variable

Presenting by

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1 PROBABILITY FUNCTION

3 VARIANCE

2 EXPECTATION OF SUM & PRODUCT

PROBABILITY FUNCTION

A **probability function** assigns probability to outcomes of a random variable.

The key rule is that the total probability of all possible outcomes must always equal 1.

The **probability function** is:

$$P(X=x)=\begin{cases}\frac{1}{6}, x=1,2,3,4,5,6\\ 0, Otherwise\end{cases}$$

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Example : Tossing a Fair Coin

Let X = outcome of a coin toss, where:

- **X=1 → Head**
- **X=0 → Tail**

The probability function (PMF) is:

$$P(X = x) = \begin{cases} 0.5\ , x = 1, Head \\ 0.5\ , x = 0, Tail \\ 0\ , Otherwise \end{cases}$$

P(X = 0) + P(X = 1) = 0.5 + 0.5 = 1

Discrete
(PMF)

A discrete random variable takes a countable number of possible values. Exp 1,2,3,4….

$$\sum_x P(X = x) = 1$$

Continuous
(PDF)

A continuous random variable takes any value within an interval .Exp 1.25,2.45…

$$\int_{\infty}^{-\infty} f(x)dx = 1$$

EXPECTATION

Expectation represents the **long-run average** or **mean** of the distribution.

Expectation of sum

The **expectation (mean)** of the sum of random variables is the **sum of their expectations**.

$$E[X+Y] = E[X] + E[Y]$$

Expectation of sum exp

Example :- Expectation of a Sum

Suppose we roll **two fair dice**, let X= first die, Y = second die.

$E[X] = E[Y] = 3.5$ (since a fair die has mean 3.5)

Expectation of the sum:

$$E[X+Y] = E[X] + E[Y] = 3.5 + 3.5 = 7$$

The sum of the two dice has an expected value of **7**.

Expectation of Product

The expectation of the product of two random variables measures the **average value of their multiplication**.

General case:

$$E[XY] \neq E[X]\,E[Y]$$

If the variables are independent:

$$E[XY] = E[X]\;E[Y]$$

Expectation of Product exp

Example :- Expectation of a Product

Suppose we toss **two fair coins**, let X = 1 if first coin is Head (0 if Tail), Y = 1 if second coin is Head (0 if Tail).

$$E[X]=E[Y]=0.5$$

If independent (coins are fair and independent):

$$E[XY] = E[X]\cdot E[Y] = 0.5 \cdot 0.5 = 0.25$$

The expected value of the product is **0.25.**

Variance of a Random Variable

Variance measures how **spread out** the values of a random variable are from their **mean expectation**.

$=$

σ

μ

λ

Binomial Distribution

$$\sigma^2 = E[X^2] - \mu^2$$

Poisson Distribution

$$\sigma^2 = E[X^2] - \mu^2 = \lambda$$

Uniform Distribution

$$\sigma^2 = \frac{(b-a)^2}{12}$$

Example : A random variable X takes values:

x	1	3
P(X = x)	0.5	0.5

Step 1: Find the expectation

$$E[\,X\,] = 1(0.5) + 3(0.5) = 2$$

Step 2: Use the variance formula

$$Var(\,X\,) = \sum(\,x - \mu\,)^2\ P(\,X = x\,)$$

$$= (\,1 - 2\,)^2(\,0.5\,) + (\,3 - 2)^2(\,0.5\,)$$

$$= (1)(0.5) + (1)(0.5)$$

$$= 1$$

Anser : Var(X)=1

Conclusion

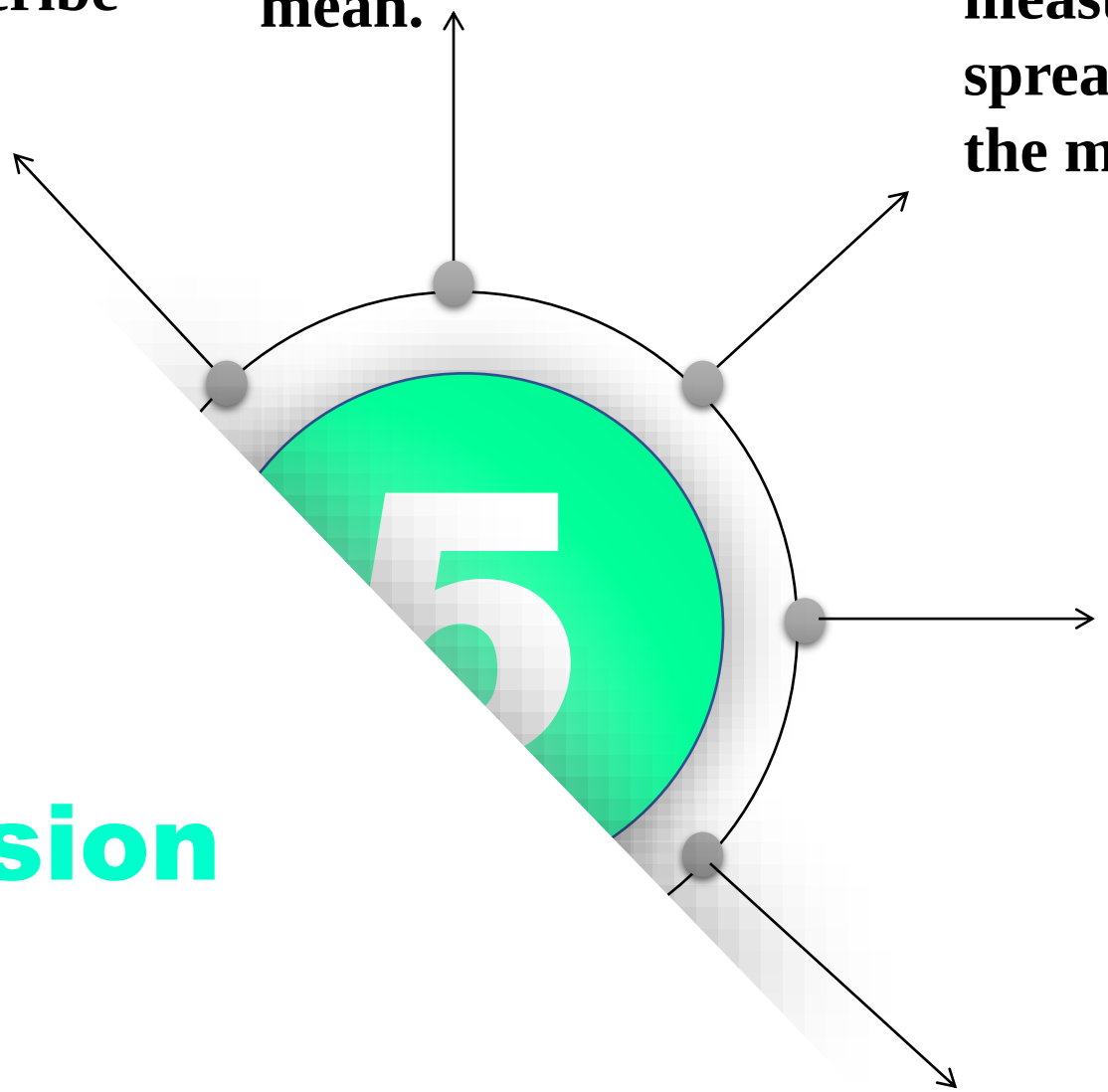
Probability functions describe outcome distributions.

Expectation is the long-term mean.

λ **Variance measures spread around the mean.**

μ **Expectation of sums adds; of products multiplies if independent.**

σ **Variance scales and adds for independent variables.**



THANK YOU